Case 7 — Ms. Anika Gupta

41 F · Slow-onset Cerebellar Ataxia · Unknown Aetiology | Wide-based gait Dysmetria Dysdiadochokinesia Dysarthria Fine motor impairment Preserved strength

Neurological Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments (Body Structure & Function)
- Truncal and gait ataxia — wide-based, unsteady gait
- Impaired postural stability and dynamic balance
- Unable to safely walk on uneven terrain or in crowds
- Difficulty using cutlery — nutrition and independence at risk
- Cannot fasten buttons or do up zips — reduced ADL independence
- Reduced community mobility and social participation

■■ RED FLAGS

- Clinical Significance
- Action if Present
- Sudden acute onset or rapid recent worsening
- Suggests cerebellar haemorrhage, posterior fossa infarct, or mass effect — a potential neurosurgical emergency
- Do not proceed with physiotherapy — immediate emergency medical referral
- Dysphagia or choking on food/liquids
- Risk of aspiration pneumonia; indicates brainstem involvement, alters prognosis
- Halt session; urgent SLT referral; document and flag to medical team

■ PATHOPHYSIOLOGY (KEY POINTS)

- The key output neurons are Purkinje cells , which provide the cerebellum's sole inhibitory output to the deep nuclei.
- In cerebellar ataxia, Purkinje cells are progressively damaged or lost — without their inhibitory influence, motor output from the deep nuclei becomes disorganised, irregular in timing, and inaccurate in force scaling.
- The cerebellar vermis and spinocerebellar tracts govern axial and gait coordination — degeneration here produces the wide-based, lurching "drunken" gait.
- Cerebellar connections to the brainstem also coordinate articulatory muscles, so Purkinje cell loss produces the characteristic scanning dysarthria heard in Anika's speech.

■ DIFFERENTIAL DIAGNOSES

See full case notes

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Test 1 — Finger-to-Nose Test (FNT)

MEASURES Limb dysmetria and intention tremor — integrity of cerebellar hemisphere coordination and sensorimotor calibration HOW TO PERFORM Seat the patient comfortably. Position yourself opposite, holding your index finger at arm's length from the patient. Ask the patient to touch the tip of their own nose with their index finger, then touch your finger tip, alternating repeatedly at a moderate pace.

Test 2 — Heel-to-Shin Test (HTS)

MEASURES Lower limb cerebellar coordination — tests cerebellar hemisphere and spinocerebellar tract function in the legs HOW TO PERFORM Ask the patient to lie supine on the plinth. Instruct them to place the heel of one foot precisely on the opposite knee.

Test 3 — Romberg Test (Eyes Open vs. Closed)

MEASURES Distinguishes cerebellar ataxia (unstable with eyes OPEN) from sensory/dorsal column ataxia (unstable only with eyes CLOSED) — identifies which sensory modality is most contributing to instability HOW TO PERFORM Ask the patient to stand with feet together, arms at sides or folded across chest. Stand directly beside the patient, close enough to catch them — never leave them unsupported.

■ TREATMENT PLAN

- Key Neurological Observations
- in cerebellar pathology — assess passively at the wrist and knee; muscles may feel "floppy" or pendular on passive movement
- Observe dysdiadochokinesia via rapid forearm pronation/supination — loss of rhythm and regularity with increasing speed confirms cerebellar involvement
- Wide-based, irregular stride length, truncal sway, tendency to veer; I also assess tandem (heel-to-toe) walking and turning — both are particularly impaired in cerebellar ataxia
- Assess sitting balance on plinth versus standing — truncal instability worsens as base of support narrows; also look for titubation (rhythmic head/trunk tremor at rest)
- Look for nystagmus, particularly gaze-evoked horizontal nystagmus — present in cerebellar or vestibulo-cerebellar dysfunction; relevant for driving and falls risk
- Dysarthria quality — cerebellar dysarthria is scanning/staccato; consistent with what is described in Anika's presentation
- Primary: Scale for the Assessment and Rating of Ataxia (SARA)
- Secondary: Berg Balance Scale (BBS)

• — provides a quantified falls risk score (≤ 45/56 = elevated risk) that is widely understood by the MDT and directly relevant to Anika's primary safety concern.
• Neurophysio Rationale
• Cerebellar rehabilitation leverages